

No-Temporal vs 主模型 (N4 Motion-Aligned RSSM)

同栈单变量：两份 config 差异仅 temporal\_fusion (None ↔ MotionAlignedRSSMFusion)

表 1 BEST 口径 (论文 / 汇报用)

| 指标                           | No-Temporal<br>(seed0) | 主模型 seed0<br>(种子对齐) |       | 主模型 seed2<br>(最高单点) |       |
|------------------------------|------------------------|---------------------|-------|---------------------|-------|
|                              |                        | N4 RSSM             | Δ     | N4 RSSM             | Δ     |
| Overall 3D moderate          | 35.59                  | 39.88               | +4.29 | 40.88               | +5.29 |
| Overall BEV moderate         | 42.23                  | 47.70               | +5.47 | 49.85               | +7.62 |
| Car 3D moderate strict       | 47.86                  | 49.98               | +2.11 | 51.35               | +3.49 |
| Truck 3D moderate strict     | 25.53                  | 30.43               | +4.90 | 30.76               | +5.23 |
| Pedestrian 3D moderate loose | 27.07                  | 28.64               | +1.57 | 31.32               | +4.25 |
| Cyclist 3D moderate loose    | 41.89                  | 50.47               | +8.58 | 50.09               | +8.20 |

表 2 窗口口径 (ep12-16 均值, 实验室选型用)

| 指标                           | No-Temporal<br>(seed0) | 主模型 seed0<br>(种子对齐) |       | 主模型 seed2<br>(最高单点) |       |
|------------------------------|------------------------|---------------------|-------|---------------------|-------|
|                              |                        | N4 RSSM             | Δ     | N4 RSSM             | Δ     |
| Overall 3D moderate          | 34.65                  | 38.39               | +3.74 | 39.40               | +4.75 |
| Overall BEV moderate         | 41.62                  | 46.52               | +4.90 | 48.33               | +6.71 |
| Car 3D moderate strict       | 46.16                  | 47.80               | +1.64 | 48.98               | +2.81 |
| Truck 3D moderate strict     | 23.48                  | 28.21               | +4.73 | 30.04               | +6.56 |
| Pedestrian 3D moderate loose | 26.67                  | 28.92               | +2.24 | 30.58               | +3.91 |
| Cyclist 3D moderate loose    | 42.28                  | 48.63               | +6.34 | 47.99               | +5.71 |

主模型三 seed (同一份 config)

| seed       | BEST Overall 3D | @ep     | BEST Overall BEV | 窗口 3D | 窗口 BEV |
|------------|-----------------|---------|------------------|-------|--------|
| seed0      | 39.88           | ep16    | 47.70            | 38.39 | 46.52  |
| seed1      | 40.51           | ep15    | 48.13            | 39.20 | 46.85  |
| seed2 (最高) | 40.88           | ep14    | 49.85            | 39.40 | 48.33  |
| 均值         | 40.42 ± 0.51    | ep14-16 | 48.56            | 39.00 | 47.23  |